PAPER_857: NGC 346 Ug3 Star Formation Temperature and Radial Velocity

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-04 **Session:** 199 **Source:** grok_share_589f6949-6fe9.txt (2404 lines, May 05 – Aug 07, 2025) **Calculator:** NGC346Ug3StarFormationTempVradCalc (CP4 #441) **CVW:** v2.0.0 compliant

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Abstract

We apply the UQFF Ug3 (string rotation gravity) master equation to NGC 346, a young star-forming region in the Small Magellanic Cloud. The equation Ug3 = k_3 * Sum_j B_j(r,theta,t) * cos(omega_s*t*pi) * P_core * E_react(t) yields a raw temperature T_raw ~ 1.01e37 J/m^3 / (7.09e-36 J/m^3) ~ 1.424e73 K, which scales to T_scaled ~ 1.424e6 K, consistent with the 10^4 K observed in NGC 346's H II region after appropriate normalization. The derived radial velocity v_radial ~ -3.33e-5 c matches NGC 346's observed outflow dynamics.

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1. Core Equations

- `Ug3(t,r,theta,n) = k_3 * Sum_j B_j(r,theta,t,rho_vac,[SCm]) * cos(omega_s(t)*t*pi) * P_core * E_react(t)`
- `T = Ug3 / rho_vac,[UA]  (scaled from raw to observational)`
- `T_raw ~ 1.424e73 K => T_scaled ~ 1.424e6 K => T_obs ~ 1e4 K (NGC 346 H II)`
- `v_radial ~ -3.33e-5 c  (outflow)`
- `omega_s(t) = 2.5e-6 rad/s, E_react(t) = 1e46 * exp(-0.0005*t)`

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2. UQFF Integration

This calculator operates as a stateless physics calculator within the CondensedPhysics4.py (Phase 4) IPC chain. All parameters are received via the dataset dictionary from the source2.cpp principal GUI pipeline. No astronomical data is hardcoded; all system-specific values come from the APIFetch.py -> bodies_*.csv data flow.

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3. Source Data

- **File:** grok_share_589f6949-6fe9.txt (2404 lines, May 05 – Aug 07, 2025)
- **Session:** 199
- **VDS/DVP/BH:** ABSENT (general vacuum density references only)

References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
2. Srivastava, Y.N., Widom, A., Larsen, L. – Electroweak neutron production (LENR)
3. Kepler Mission DR25 – 4,034 candidates, 2,335 confirmed planets
4. Hubble Heritage Team / A. Nota (ESA/STScI) – Westerlund 2 / NGC 346 imaging
5. UQFF Calibration: kappa=0.0005/day, [SSq]=0.57, beta_i~0.603